

MATH503 Mathematics for Computing

Week 10

Matrix definition and operations

Transformation matrices

Housekeeping: Assessments

➤ yimaths:

Catch-up if you are behind

➤ ortfo io activities:

Check your grades on Canvas

Two more activities in Week 10 and Week 11

➤ Exam information:

Link: <https://canvas.aut.ac.nz/courses/7554pages/ecms-examinations>

More information next week

MATH503 Mathematics for Computing

6.1 Matrix definition and operations

Matrix dimension

Notation

Equal matrices

Special matrices

Transpose

Matrix addition

Scalar multiplication

Matrix multiplication

Why matrices?

Computer science relies heavily on matrices and their inverses:

- ▶ Projection of a 3-dim image onto a 2-dim screen
- ▶ Encryption of message codes
- ▶ Page rank algorithms in search engines
- ▶ Recording data

Matrix dimension

Reading: Tan p102 - 109, p115 - 121 or LGR p86 - 103

A **matrix** is a rectangular array of values

It has **dimension** or **order** $m \times n$ if it contains m rows and n columns

Note that the number of rows is specified first

If $m = n$ it is called a **square** matrix

If $m = 1$ it is called a **row matrix**

If $n = 1$ it is called a **column matrix**

Example 1

$$A = \begin{pmatrix} 12 & 7 & -1 \\ 2 & 0 & \frac{3}{2} \end{pmatrix}$$

$$B = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$$

$$C = \begin{pmatrix} 7 \\ -3 \\ 2 \end{pmatrix}$$

A is a 2×3 matrix

B is a square 2×2 matrix

C is a column 3×1 matrix

Notation

A general $m \times n$ matrix A is written as follows:

$$A = \begin{pmatrix} a_{1\,1} & \cdots & a_{1\,n} \\ \vdots & & \vdots \\ a_{m\,1} & \cdots & a_{m\,n} \end{pmatrix} \text{ or } A = [a_{ij}]$$

Equal matrices

Two matrices A and B are said to be **equal**

- ▶ if they have the same dimension
- ▶ if they have the same corresponding entries
that is, the i,j -entry of A is the same as the i,j -entry of B

Example 2

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \qquad B = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 4 & 0 \end{pmatrix}$$

$A \neq B$. Why?

Special matrices

Zero matrix

Example 3

$$O_{2 \times 2} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$O_{2 \times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Diagonal matrix

Example 4

$$A = \begin{pmatrix} -3 & 0 \\ 0 & 5 \end{pmatrix}$$

$$B = \begin{pmatrix} -0.5 & 0 & 0 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & 7 \end{pmatrix}$$

Identity matrix

Example 5

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Transpose

The **transpose** of an $m \times n$ matrix A

is the $n \times m$ matrix A^T

obtained by interchanging the rows and columns of A

so the i, j -entry of A becomes the j, i -entry of A^T

Example 6

$$P = \begin{pmatrix} 4 & -3 & 2 \\ 0 & 1 & -1 \end{pmatrix} \quad P^T = \begin{pmatrix} & \\ & \\ & \end{pmatrix}$$

Matrix addition

Example 7

$$A + B = \begin{pmatrix} 12 & 7 & -1 \\ 2 & 0 & \frac{3}{2} \end{pmatrix} + \begin{pmatrix} 3 & 0 & -4 \\ \frac{1}{3} & 9 & -1 \end{pmatrix} = \begin{pmatrix} & & \\ & & \end{pmatrix}$$

Scalar multiplication

Example 8

$$2A = 2 \begin{pmatrix} 12 & 7 & -1 \\ 2 & 0 & \frac{3}{2} \end{pmatrix} = \begin{pmatrix} 24 & 14 & -2 \\ 4 & 0 & 3 \end{pmatrix}.$$

Matrix multiplication

A is a $m \times n$ matrix and B is a $n \times p$ matrix

Then the matrix product AB is a $m \times p$ matrix

Calculate the entries in the product as shown in the diagram

$$AB = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ \boxed{a_{i1} \quad \cdots \quad a_{in}} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} b_{11} & \cdots & \boxed{b_{1j}} & \cdots & b_{1p} \\ \vdots & & \vdots & & \vdots \\ b_{n1} & \cdots & \boxed{b_{nj}} & \cdots & b_{np} \end{pmatrix} = \begin{pmatrix} c_{11} & \cdots & c_{1p} \\ \vdots & & \vdots \\ c_{m1} & \boxed{c_{ij}} & c_{mp} \end{pmatrix}$$

where $c_{ij} = a_{i1}b_{1j} + \cdots + a_{in}b_{nj}$

Example 9

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & 3 \\ 5 & 7 \end{pmatrix}$$

A is a 2×2 matrix and B is a 2×2 matrix.

The product AB will be a 2×2 matrix.

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 5 & 7 \end{pmatrix} =$$

Note that usually $AB \neq BA$.

$$BA = \begin{pmatrix} 1 & 3 \\ 5 & 7 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} =$$

Example 10

$$A = \begin{pmatrix} 12 & 7 & -1 \\ 2 & 0 & -3 \end{pmatrix}, \quad B = \begin{pmatrix} 1 \\ 5 \\ -1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & -1 & 0 & 1 \\ 2 & 0 & 0 & 1 \\ 6 & -4 & 0 & 1 \end{pmatrix}$$

Which of the following products are possible?

AB

BA

AC

CA

BC

CB

Calculate them.

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6.2 Inverse of a matrix

Definition of inverse matrix

Reading: LGR p580 – 587, p107 – 111

A square $n \times n$ matrix A has an inverse if and only if there exists another $n \times n$ matrix B such that

$$AB = BA = I_n.$$

We call B the **inverse** of A and write $A^{-1} = B$.

Example

Example 1

The matrix $A = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$ has the inverse $A^{-1} = \begin{pmatrix} -5 & 3 \\ 2 & -1 \end{pmatrix}$

To show that this is true, calculate the following two matrix products:

$$\begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} -5 & 3 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

and

$$\begin{pmatrix} -5 & 3 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Inverse matrix for a 2×2 matrix A

The following procedure is used to calculate the inverse of a 2×2 matrix A

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

First find the number $a_{11}a_{22} - a_{12}a_{21}$

This number is called the **determinant** $\det(A)$ of the matrix A

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$$

Example 2

Use the procedure to find the inverse matrix for $A = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$

$$\det(A) =$$

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix} =$$

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6.3 Transformation with matrices

Matrices and transformation

Reading: MyiMaths/Classic MyiMaths/IGCSE /Matrices

A point in the plane is represented by the column matrix $\begin{pmatrix} x \\ y \end{pmatrix}$

Matrices can be used to change the position of the point

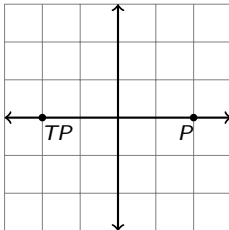
Example 1

$P = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$ is a point on the x -axis

Let $T = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

Then $TP = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$

The point P is reflected in the y -axis and TP is on the negative x -axis



Matrices and transformation

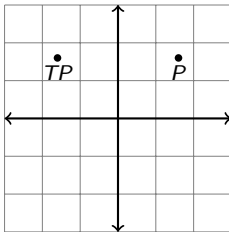
Example 1 (continued)

$P = \begin{pmatrix} a \\ b \end{pmatrix}$ is a point in the first quadrant

Let $T = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

Then $TP = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -a \\ b \end{pmatrix}$

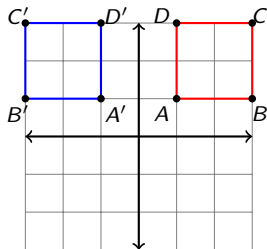
The point P is reflected in the y -axis and TP is in the second quadrant



Matrices and transformation

A geometric figure with straight line sides can be transformed by multiplying the vertices with the transformation matrix

Example 1 (continued)



$$T(A \ B \ C \ D) = (A' \ B' \ C' \ D')$$

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 & 3 & 1 \\ 1 & 1 & 3 & 3 \end{pmatrix} = \begin{pmatrix} -1 & -3 & -3 & -1 \\ 1 & 1 & 3 & 3 \end{pmatrix}$$

Reflections

The following are **reflection matrices**:

► $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ reflects the point in the line $x = 0$ (y -axis)

► $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ reflects the point in the line $y = 0$ (x -axis)

► $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ reflects the point in the line $y = x$

► $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ reflects the point in the line $y = -x$

Reflection in the line $y = -x$

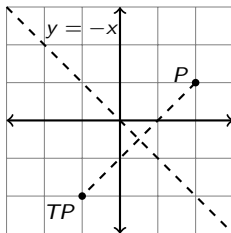
Example 2

$P = \begin{pmatrix} a \\ b \end{pmatrix}$ is a point in the first quadrant

Let $T = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

Then $TP = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -b \\ -a \end{pmatrix}$

The point P is reflected in the line $y = -x$ and TP is in the third quadrant



Rotations

Some rotations with centre of rotation at the origin can be represented with rotation matrices.

The following are **rotation matrices**:

- ▶ $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ rotates a point clockwise round the origin through 90°
- ▶ $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ rotates a point anticlockwise round the origin through 90°
- ▶ $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ rotates a point round the origin through 180°

Rotation clockwise through 90°

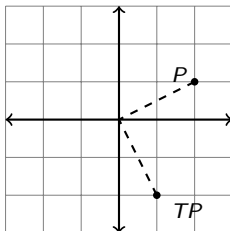
Example 3

$P = \begin{pmatrix} a \\ b \end{pmatrix}$ is a point in the first quadrant

Let $T = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

Then $TP = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} b \\ -a \end{pmatrix}$

The point P is rotated clockwise round the origin through 90° and TP is in the fourth quadrant



Enlargements and stretches

The **enlargement matrix** $\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$ (k a positive number) produces an enlargement with scale factor k .

The **stretch matrix** $\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}$ (k a positive number) is a stretch in the x -direction with scale factor k .

The **stretch matrix** $\begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$ (k a positive number) is a stretch in the y -direction with scale factor k .

Stretch in x -direction with scale factor 2

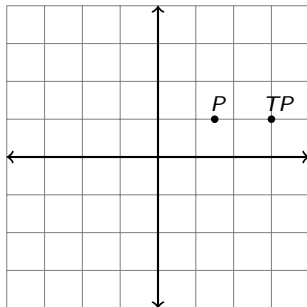
Example 4

$P = \begin{pmatrix} a \\ b \end{pmatrix}$ is a point in the first quadrant

Let $T = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$

Then $TP = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 2a \\ b \end{pmatrix}$

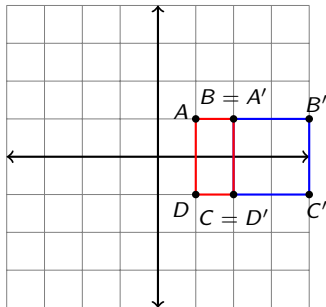
The point P is stretched by a factor 2 in the x -direction and TP is in the first quadrant



Stretch in x -direction with scale factor 2

A geometric figure with straight line sides can be transformed by multiplying the vertices with the transformation matrix

Example 4 (continued)



$$T(A \ B \ C \ D) = (A' \ B' \ C' \ D')$$

$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 2 & 1 \\ 1 & 1 & -1 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 4 & 4 & 2 \\ 1 & 1 & -1 & -1 \end{pmatrix}$$

Shear transformation

The **shear matrix** $\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$ produces a shear in the x -direction with shear factor k .

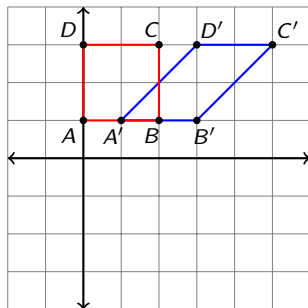
The **shear matrix** $\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$ produces a shear in the y -direction with shear factor k .

Shear in x -direction with shear factor 1

A geometric figure with straight line sides can be transformed by multiplying the vertices with the transformation matrix

Example 5

$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is the shear matrix with shear factor 1 in the x -direction.



$$T(A \ B \ C \ D) = (A' \ B' \ C' \ D')$$

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 2 & 2 & 0 \\ 1 & 1 & 3 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 3 & 5 & 3 \\ 1 & 1 & 3 & 3 \end{pmatrix}$$

Determinant and inverse of a transformation matrix

Inverse of a transformation matrix

The inverse matrix T^{-1} is also a transformation matrix.

The inverse T^{-1} will reverse or undo the effect of T .

Determinant of a transformation matrix

The $\det(T)$ of a transformation matrix gives the degree to which the area of an image is increased or decreased:

$$\text{Area of image} = \det(T) \times \text{Area of original}$$

Use the absolute value of the $\det(A)$.

If the determinant is negative, the transformation includes a reflection.